

Local fault property modeling



Full-fault property sampling

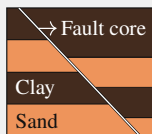


Field-scale CO₂ storage simulation

■ Geologic uncertainty

A throw-window
geologic model

Vary structural and
stratigraphic parameters



z_f
 T_i
 $V_{cl,sand}$
 $V_{cl,clay}$

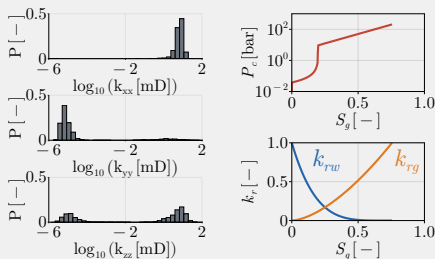
162 geologic
scenarios

■ Stochastic fault-core architectures

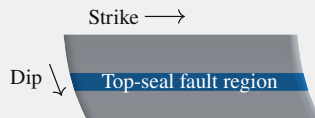
$N = 2,000$ per geologic scenario



■ Physics-based fault-property upscaling



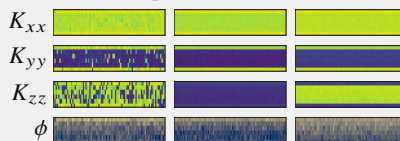
■ 3D fault geometry



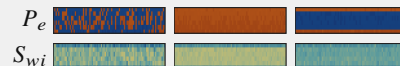
6 windows along dip $\times$ 87 slices along strike

■ Fault property realizations

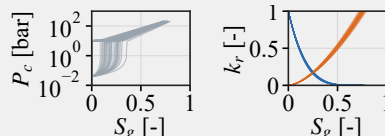
Example K and ϕ fields



Multiphase flow properties



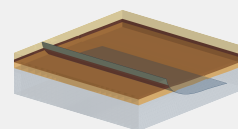
Window 4: 87 along-strike P_c and k_r curves



12 stochastic fields + 3 benchmarks
per geologic scenario

■ Offshore Texas reservoir model

~ 2 million gridblocks



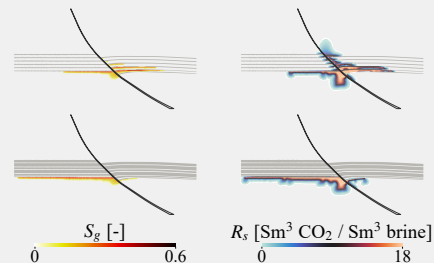
Overburden

Top seal

Storage reservoir

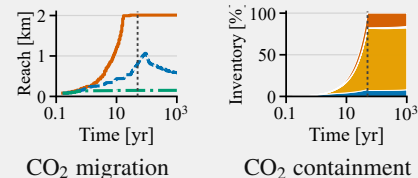
Underburden

■ CO₂ migration along and across the fault



■ Assessment and targeted enrichment

2,430 Phase-1 reservoir simulations



End-to-end uncertainty quantification workflow for geologic CO₂ storage in faulted reservoirs. PREDICT generates stochastic local fault properties, and conditionally independent sampling constructs spatially heterogeneous full-fault fields for JutulDarcy simulation of CO₂ migration and containment.